Assignment 5

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Problem 1

Consider the op-amp circuit below. Let $v_1(t)$, $v_2(t)$, and $v_3(t)$ denote the voltages across the three capacitors, and $v_0(t)$ denote the *input* voltage.

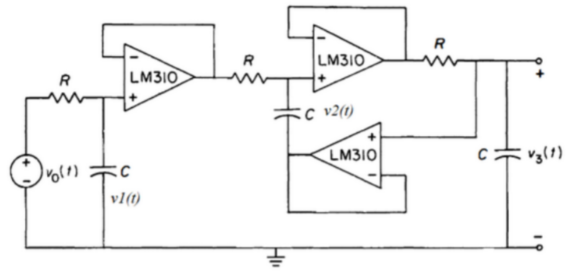

Figure 1: Figure for Problem 1

The circuit equations are as follows:

$$\dot{v}_1(t) + \frac{1}{RC}\ (\ v_1(t) - v_0(t)\ ) = 0$$
$$\dot{v}_2(t) + \frac{1}{RC}\ (\ v_3(t) + v_2(t) - v_1(t)\ ) = 0$$
$$\dot{v}_3(t) + \frac{1}{RC}\ (\ v_3(t) - v_2(t)\ ) = 0$$

(a) Suppose you have access only to observing the voltage $v_3(t)$. Determine whether the system is observable or not.

(b) Suppose you now observe the voltage $v_3(t) + v_2(t)$. Check if this system is observable.

1(a)

```
clc;clear;close all; syms R C;

A = -(1/(R*C))*[1 0 0; -1 1 1; 0 -1 1];
B = [-1;0;0];
Cmat = [0 0 1];     % named Cmat to avoid clash with symbolic C

Ob = [Cmat;Cmat*A;Cmat*A^2]
```

```
Ob =
```

$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & \frac{1}{C\,R} & -\frac{1}{C\,R} \\ \frac{1}{C^2\,R^2} & -\frac{2}{C^2\,R^2} & 0 \end{bmatrix}$$

```
Ob_reduced = rref(Ob)
```

```
Ob_reduced =
```

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

```
is_observable = (rank(Ob) == size(A,1))
```

```
is_observable = logical
   1
```

1(b)

```
clc;clear;close all; syms R C;

% same A & B as in (a)
A = -(1/(R*C))*[1 0 0; -1 1 1; 0 -1 1];
B = [-1;0;0];
Cmat = [0 1 1];

Ob = [Cmat;Cmat*A;Cmat*A^2]
```

Ob =

$$\begin{bmatrix} 0 & 1 & 1 \\ \frac{1}{C\,R} & 0 & -\frac{2}{C\,R} \\ -\frac{1}{C^2\,R^2} & -\frac{2}{C^2\,R^2} & \frac{2}{C^2\,R^2} \end{bmatrix}$$

```
Ob_reduced = rref(Ob)
```

Ob_reduced =

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

```
is_observable = (rank(Ob) == size(A,1))
```

```
is_observable = logical
   1
```

Problem 2

A system can be described as

$$\dot{X} = AX + BU$$
$$Y = CX,$$

where $A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 3\omega_0^2 & 0 & 0 & 2\omega_0 r_0 \\ 0 & 0 & 0 & 1 \\ 0 & -\frac{2\omega_0}{r_0} & 0 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 0 \\ 0 & 2 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$, and C will be specified for parts (b) thru (e).

(a) Check whether the system is controllable

(b) Suppose you observe both x_1 and x_3. Is the system observable?

(c) Suppose you observe the sum $x_1 + x_3$. Is the system observable?

(d) Suppose you only observe x_1. Is the system observable?

(e) Suppose you only observe x_3. Is the system observable?

2(a)

```
clc;clear;close all; syms omega_0 r_0;

A = [0 1 0 0; 3*omega_0^2 0 0 2*omega_0*r_0; 0 0 0 1; 0 -(2*omega_0)/r_0 0 0]
```

A =

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 3\,{\omega_0}^2 & 0 & 0 & 2\,\omega_0\,r_0 \\ 0 & 0 & 0 & 1 \\ 0 & -\frac{2\,\omega_0}{r_0} & 0 & 0 \end{bmatrix}$$

```
B = [0 0; 0 2; 0 0; 0 0];

CO = [B A*B A^2*B A^3*B]
```

CO =

$$\begin{bmatrix} 0 & 0 & 0 & 2 & 0 & 0 & 0 & -2\,{\omega_0}^2 \\ 0 & 2 & 0 & 0 & 0 & -2\,{\omega_0}^2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{4\,\omega_0}{r_0} & 0 & 0 \\ 0 & 0 & 0 & -\frac{4\,\omega_0}{r_0} & 0 & 0 & 0 & \frac{4\,{\omega_0}^3}{r_0} \end{bmatrix}$$

```
CO_reduced = rref(CO)
```

CO_reduced =

$$\begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & -{\omega_0}^2 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

```
is_controllable = (rank(CO) == size(A,1))
```

```
is_controllable = logical
   0
```

2(b)

```
clc;clear;close all; syms omega_0 r_0;

A = [0 1 0 0; 3*omega_0^2 0 0 2*omega_0*r_0; 0 0 0 1; 0 -(2*omega_0)/r_0 0 0];
B = [0 0; 0 2; 0 0; 0 0];
Cb = [1 0 0 0; 0 0 1 0];

OBb = [Cb;Cb*A;Cb*A^2;Cb*A^3]
```

OBb =

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 3\,{\omega_0}^2 & 0 & 0 & 2\,\omega_0\,r_0 \\ 0 & -\frac{2\,\omega_0}{{r_0}^2} & 0 & 0 \\ 0 & -{\omega_0}^2 & 0 & 0 \\ -\frac{6\,{\omega_0}^3}{r_0} & 0 & 0 & -4\,{\omega_0}^2 \end{bmatrix}$$

```
rref(OBb)
```

ans =

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

```
is_observable = (rank(OBb) == size(A,1))
```

```
is_observable = logical
   1
```

2(c)

```
clc;clear;close all; syms omega_0 r_0;

A = [0 1 0 0; 3*omega_0^2 0 0 2*omega_0*r_0; 0 0 0 1; 0 -(2*omega_0)/r_0 0 0];
B = [0 0; 0 2; 0 0; 0 0];
Cc = [1 0 1 0]
```

```
Cc = 1x4
     1     0     1     0
```

```
OBc = [Cc;Cc*A;Cc*A^2;Cc*A^3]
```

OBc =

$$\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 3\,{\omega_0}^2 & -\frac{2\,\omega_0}{r_0} & 0 & 2\,\omega_0\,r_0 \\ -\frac{6\,{\omega_0}^3}{r_0} & -{\omega_0}^2 & 0 & -4\,{\omega_0}^2 \end{bmatrix}$$

```
rref(OBc)
```

ans =

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

```
is_observable = (rank(OBc) == size(A,1))
```

```
is_observable = logical
   1
```

2(d)

```
clc;clear;close all; syms omega_0 r_0;

A = [0 1 0 0; 3*omega_0^2 0 0 2*omega_0*r_0; 0 0 0 1; 0 -(2*omega_0)/r_0 0 0];
B = [0 0; 0 2; 0 0; 0 0];
Cd = [1 0 0 0]
```

```
Cd = 1x4
     1     0     0     0
```

```
OBd = [Cd;Cd*A;Cd*A^2;Cd*A^3]
```

OBd =

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 3\,{\omega_0}^2 & 0 & 0 & 2\,\omega_0\,r_0 \\ 0 & -{\omega_0}^2 & 0 & 0 \end{bmatrix}$$

```
rref(OBd)
```

ans =

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

```
is_observable = (rank(OBd) == size(A,1))
```

```
is_observable = logical
   0
```

2(e)

```
clc;clear;close all; syms omega_0 r_0;

A = [0 1 0 0; 3*omega_0^2 0 0 2*omega_0*r_0; 0 0 0 1; 0 -(2*omega_0)/r_0 0 0];
B = [0 0; 0 2; 0 0; 0 0];
Ce = [0 0 1 0]
```

```
Ce = 1x4
     0     0     1     0
```

```
OBe = [Ce;Ce*A;Ce*A^2;Ce*A^3]
```

OBe =

$$\begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & -\frac{2\,\omega_0}{r_0} & 0 & 0 \\ -\frac{6\,{\omega_0}^3}{r_0} & 0 & 0 & -4\,{\omega_0}^2 \end{bmatrix}$$

```
rref(OBe)
```

ans =

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

```
is_observable = (rank(OBe) == size(A,1))
```

```
is_observable = logical
   1
```